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SEMINAR REPORT

Seminar (24UCYPC201), S.Y. B.Tech Cyber Security

Sr.	Index	Page No
1	Introduction and Problem statement	5
2	Objectives	6
3	methodology and System design	7
4	Implementation	10
5	Results and security analysis	11
6	Conclusion and Future scope	12

A Project Report on

CYBER LAW AND FORENSICS

PS Code:
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Complexity: Intermediate

Submitted in partial fulfilment of the requirements of the course
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Department of Cyber Security
School of Engineering and Technology
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CERTIFICATE

This is to certify that the Seminar report entitled **Cyber Law and Forensics** (PS Code:) submitted by **Aniket Sanjiv Jadhav** PRN : **2125UCSM1013** is a bonafide record of the laboratory project work carried out by him/her under my guidance, in partial fulfilment of the requirements of the course SEMINAR (24UCYPC201) Laboratory, S.Y. B.Tech Cyber Security, during the Academic Year 2026-2027 at the Department of Cyber Security, School of Engineering and Technology, Sanjivani University, Kopargaon.

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CYBER LAW AND FORENSICS

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Abstract

The rapid growth of digital technology has increased the occurrence of cybercrimes such as unauthorized access, phishing, identity theft, malware attacks, and online fraud. Cyber law provides the legal framework for dealing with activities and offences involving information technology, while digital forensics provides methods for collecting, preserving, examining, and analyzing digital evidence. This seminar studies the relationship between cyber law and digital forensics and explains the role of digital evidence in cybercrime investigation. A systematic forensic methodology involving evidence identification, preservation, acquisition, integrity verification, examination, analysis, and reporting is discussed. Tools such as Autopsy, FTK Imager, and Wireshark are considered for practical investigation. SHA-256 hashing is used as an example of an integrity-verification technique. The study shows that combining legal requirements with systematic forensic procedures improves the reliability and usefulness of digital evidence during cybercrime investigations.

Keywords: Cyber Law, Digital Forensics, Cybercrime, Digital Evidence, Chain of Custody

1. Introduction and Problem Statement

The rapid development of information technology has significantly changed the way people communicate, work, study, conduct business, and access services. Computers, smartphones, internet services, social media, online banking, cloud platforms, and digital payment systems have become an important part of everyday life. However, the increasing dependence on digital technologies has also resulted in new forms of criminal activity known as cybercrime.

Cybercrime involves the use of computers, networks, digital devices, or online services to commit illegal activities. Examples include unauthorized access, phishing, identity theft, financial fraud, malware attacks, ransomware, data theft, cyberstalking, and denial-of-service attacks. Such crimes can be difficult to investigate because the attacker may operate remotely, use multiple systems, hide their identity, or delete and modify digital information.

Cyber law provides rules and legal principles for activities involving computers, networks, electronic communication, and cyberspace. It helps establish the legal framework for dealing with cyber offences and electronic records. In India, the Information Technology Act, 2000 is an important part of the legal framework related to information technology and cyber offences.

Digital forensics is the technical process of identifying, collecting, preserving, examining, analyzing, and reporting digital information that may be relevant to an investigation. Digital evidence can be obtained from computers, mobile phones, storage devices, networks, emails, cloud services, system logs, and applications.

Cyber law and digital forensics are closely related. Cyber law provides the legal context for cybercrime investigation, while digital forensics provides technical methods for discovering and analyzing evidence. A forensic investigation must therefore consider both technical procedures and applicable legal requirements.

Problem Statement

The increasing number and complexity of cyber incidents create difficulties in identifying, preserving, analyzing, and presenting digital evidence. Digital evidence can be easily modified, deleted, encrypted, or transferred between different systems. Investigators therefore require a systematic methodology that preserves evidence integrity and supports reliable investigation.

The problem addressed by this seminar is to understand **how cyber law and digital forensic techniques can be combined to investigate cyber incidents, preserve digital evidence, maintain its integrity, and support authorized legal proceedings.**

Existing digital forensic tools such as Autopsy, FTK Imager, and Wireshark provide useful technical capabilities, but tools alone are insufficient. Proper evidence preservation, documentation, chain of custody, integrity verification, and legal awareness are also necessary for a reliable investigation.

2. Objectives

The objectives of this seminar are:

1. **To understand** the fundamental concepts of cyber law and its importance in addressing cybercrime.
2. **To study** the digital forensic investigation process, including identification, preservation, acquisition, examination, analysis, and reporting of digital evidence.
3. **To examine** the importance of evidence integrity and chain of custody in maintaining the reliability of digital evidence.
4. **To study** commonly used digital forensic tools and understand how forensic findings can support cybercrime investigation.

3. Methodology and System Design

The proposed methodology combines legal considerations with a systematic digital forensic investigation process. The investigation begins when a cyber incident is identified and potential sources of digital evidence are located.

Figure 3.1: Digital Forensic Investigation Process

Cyber Incident → Evidence Identification → Preservation → Acquisition → Hash Verification → Examination → Analysis → Reporting

Evidence Identification

Potential sources of evidence such as computers, storage devices, mobile devices, network captures, logs, and emails are identified.

Evidence Preservation

The identified evidence is protected from accidental or unauthorized modification. The original evidence should be handled carefully and forensic copies should be used for analysis wherever appropriate.

Evidence Acquisition

A forensic copy or image of the relevant storage media is created. The acquired copy can then be examined without unnecessarily modifying the original evidence.

Integrity Verification

A cryptographic hash such as SHA-256 can be calculated for digital evidence. The hash value can be compared during verification to help identify whether the evidence has changed.

Hash Process:

Digital Evidence → SHA-256 → Hash Value

SHA-256 produces a 256-bit hash value. For this seminar, SHA-256 is considered because it is widely used for integrity verification.

Examination and Analysis

The forensic copy is examined using appropriate tools. Relevant files, metadata, browser information, logs, deleted artifacts, and other evidence are identified. The information is then analyzed and correlated to understand the incident.

Reporting

The findings are documented along with the evidence examined, methodology followed, observations, and conclusions.

Threat Model

The investigation considers threats such as evidence modification, deletion, unauthorized access, log manipulation, encryption, and anti-forensic activities. Hash verification, controlled evidence handling, documentation, and forensic analysis help reduce these risks.

4. Implementation

The practical implementation can be demonstrated using commonly available forensic tools.

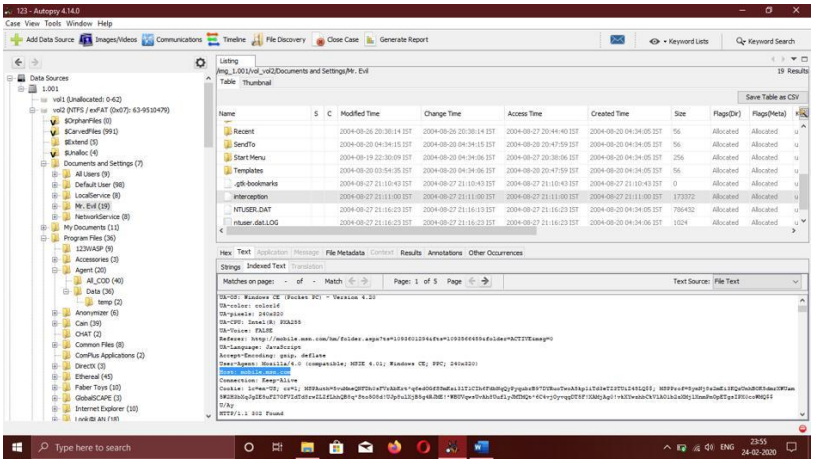
Tools and Technologies

Tool/Technology	Purpose
Autopsy	Digital forensic examination
FTK Imager	Evidence acquisition and preview
Wireshark	Network traffic analysis
Python	Hash verification demonstration
SHA-256	Evidence integrity verification

Autopsy

Autopsy is a digital forensic investigation platform used to examine forensic images and identify digital artifacts. It can assist in file-system analysis, keyword searching, metadata examination, deleted-file analysis, and timeline investigation.

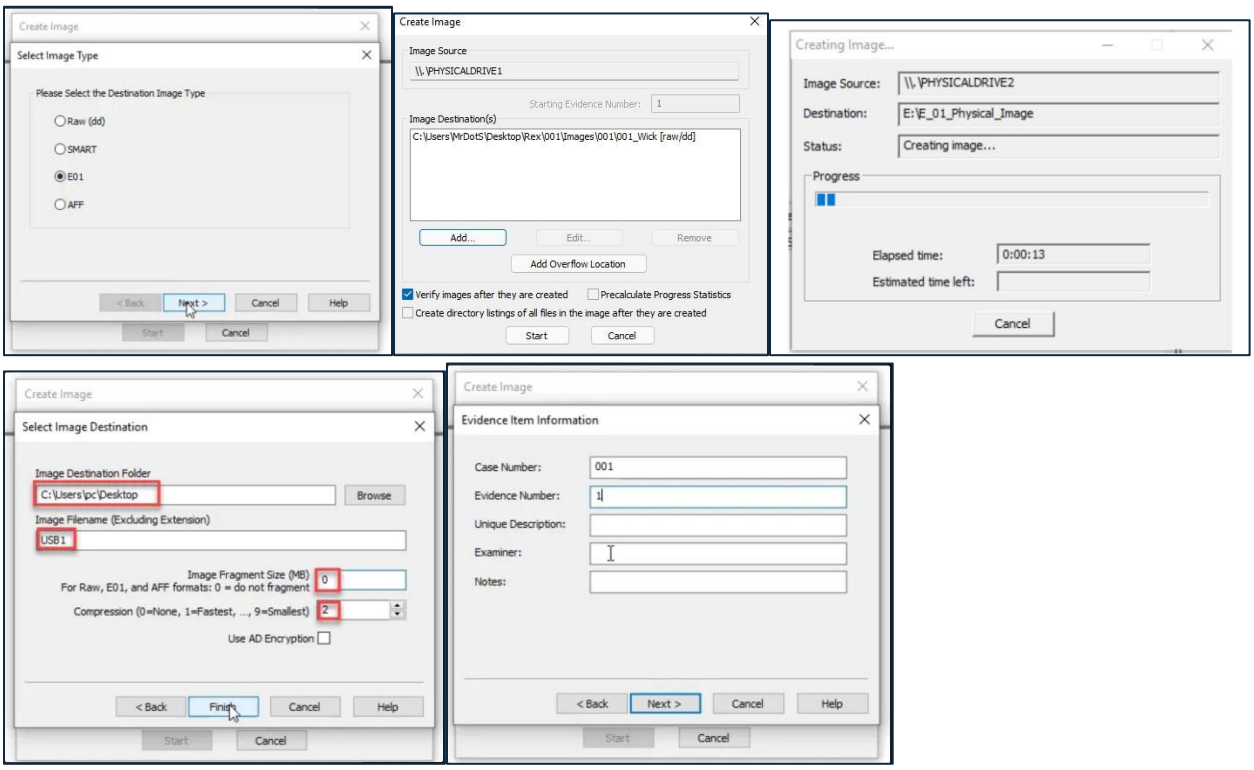
Figure 4.1: Autopsy Forensic Analysis



FTK Imager

FTK Imager can be used to preview evidence and create forensic images. The investigator can select the evidence source, create an image, record evidence information, and verify the resulting data.

Figure 4.2: FTK Imager Evidence Acquisition



Wireshark

Wireshark is used to capture and analyze network packets. It can provide information about IP addresses, protocols, ports, timestamps, and network communication.

SHA-256 Verification

A short Python implementation can demonstrate the calculation of a SHA-256 hash:

```
import hashlib
def calculate_hash(file_name):
    h = hashlib.sha256()
    with open(file_name, "rb") as f:
        for block in iter(lambda: f.read(4096), b""):
            h.update(block)
    return h.hexdigest()

print(calculate_hash("evidence.img"))
```

The hash can be compared later to verify evidence integrity.

5. Results and Security Analysis

The methodology provides a structured process for handling digital evidence from identification through reporting. The use of forensic tools makes it possible to examine storage media, files, network traffic, and other digital artifacts.

Table 5.1: Test Cases

Test ID	Input / Action	Expected Output	Actual Output	Status
TC01	Create forensic case and add evidence	Evidence should be loaded	Evidence loaded successfully	Pass
TC02	Calculate SHA-256 hash	Hash value generated	Hash value generated	Pass
TC03	Analyze network/file artifacts	Relevant artifacts displayed	Relevant artifacts displayed	Pass

Security Analysis

Evidence Integrity:

Hashing helps identify changes made to digital evidence.

Evidence Preservation:

Creating forensic copies reduces the need to work directly on original evidence.

Chain of Custody:

Documentation of evidence handling provides accountability throughout the investigation.

Incident Reconstruction:

Combining information from files, logs, timestamps, and network activity can help establish the

Limitations

Digital forensic investigation can be affected by encryption, large amounts of data, deleted information, cloud storage, changing technologies, and anti-forensic techniques. The availability and quality of evidence can also affect the final investigation results.

GitHub Link -

6. Conclusion and Future Scope

Cyber law and digital forensics are important components of modern cybercrime investigation. Cyber law provides the legal framework, while digital forensics provides the technical procedures required to investigate digital incidents.

This seminar studied the process of identifying, preserving, acquiring, examining, and analyzing digital evidence. The importance of hash verification and chain of custody was also discussed. Tools such as Autopsy, FTK Imager, and Wireshark can support different stages of a forensic investigation.

The study shows that reliable digital investigation requires both technical procedures and appropriate legal considerations. Proper evidence handling and documentation improve the integrity and reliability of forensic findings.

Future work can focus on AI-assisted forensic analysis, cloud forensics, mobile-device forensics, and automated correlation of evidence from multiple digital sources.

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